

Project Instructions — Stage 05: Data Storage (preview)

All paths below are inside `project/`.

Today's Project Contribution:

Today you'll complete a piece of your full data project.

This task aligns with the **Data Storage** lifecycle stage, where you will:

- Save and reload raw data reproducibly using environment-driven paths.
- Add storage conventions and documentation to your repo.

By the end of this assignment, your project should include the elements listed below.

Deliverable Options (from Stage Scaffold)

Required:

- `data/raw/` and `data/processed/` folder structure.
- At least one saved raw data file (e.g., `.csv` or `.parquet`).
- Add a **Data Storage** section to `README.md` explaining:
 - data folder structure,
 - file formats used, and
 - how your code reads from these files.
- Use of `.env` + secure file path handling (e.g., via `os.getenv(...)`) — especially if loading from protected locations or environment variables set earlier.
- Add a cell to `notebooks/project_pipeline.ipynb` that runs what you built this stage, then run the notebook top to bottom to confirm everything you have built so far still works.

How This Fits Into Your Final Project:

Your work today ensures anyone can **recreate your data state** by running your IO code, unlocking reliable preprocessing and modeling later.

Before next class:

- Save files in `data/raw/` and `data/processed/` as appropriate.
- Commit and push changes to your GitHub repo (never commit `.env`).
- Review assumptions, risks, and validation notes in your notebook/README.

Explicit Chain:

In your **homework**, you produced **CSV + Parquet saves, reloads, and IO utilities with env-driven paths**.

Now, you will adapt that work to formalize your project's storage layer and documentation so teammates can run it without path edits.

By the end of the course: your full project will follow the complete lifecycle.
